Empathic_AI Chatbot

Aziz Mohammed
COMP741 - Practical AI
Professor Sabin
28 April 2025

Goals: *Create a sarcasm detecting emotional therapy chatbot to brighten your day!*

Objectives:

AI Approaches:

- Transformer-based text classification and generation
- Parameter-efficient fine-tuning via LoRA
- Prompt engineering for context-aware responses

Systems Experimented With:

- DistilBERT — sarcasm classifier
- Qwen2.5-3B — empathetic response generator
- Gradio — interactive chat interface

Experiments:

- Fine-tuned DistilBERT on 50k Reddit comments → evaluated with accuracy + F1
- Fine-tuned Qwen2.5-3B on 2,350 therapy conversations → monitored training loss

Source	Role	Credibility
HuggingFace Transformers	Model loading + training	Industry standard, peer reviewed
PEFT / LoRA	Fine-tuning framework	Hu et al. 2021, widely cited
mental-health-chat-dataset	Therapist training data	Real anonymized therapy Q&A
daniel2588/sarcasm	Sarcasm training data	1M+ rows, noted label noise
Google Colab T4 GPU	Compute resource	Google Research infrastructure

Theoretical Background:

Transformer Architecture:

- Neural network that processes text by weighing relationships between all words simultaneously
- Enables deep contextual understanding of language

Fine-tuning:

- Adapting a pretrained model to a specific task using domain data
- More efficient than training from scratch — preserves existing language knowledge

LoRA (Low-Rank Adaptation)

- Fine-tuning technique that injects small trainable matrices into frozen model layers
- Reduces trainable parameters by ~99%, making large model fine-tuning feasible on limited hardware

Text Classification

- Assigning predefined labels to input text using learned patterns
- Enables automated categorization of language at scale

Theoretical Background Continued:

Neural-based:

- Systems that learn patterns directly from data through training
- Used to train both the sarcasm classifier and the empathetic response model

Reasoning-based:

- Systems that follow explicit human-defined rules and logic
- Used to route user input to the correct response strategy based on sarcasm detection output

Combined Approach:

- Neural models handle language understanding; logic layer connects their outputs
- Produces responses that are both linguistically natural and contextually appropriate

Development Sources:

HuggingFace Transformers:
- Access thousands of pretrained models in a few lines of code
- Handles tokenization, model loading, and training

PEFT:
- Implements LoRA fine-tuning
- Freezes base model, injects small trainable adapters

TRL:
- Training loop built for language model fine-tuning
- Handles batching, checkpointing automatically

Bitsandbytes:
- Compresses model from ~12GB to ~4GB via 4-bit quantization
- Makes 3B model fit within T4's 15GB VRAM

Google Colab T4 GPU:
- Free cloud GPU with 15GB dedicated memory
- Runs training ~100x faster than CPU

Pipeline Diagram:

```
[User Input]
    ↓
[DistilBERT — Sarcasm Classifier]
    ↓
[Sarcastic?]
  Yes → "User is frustrated, respond gently"
  No  → "Respond as supportive therapist"
    ↓
[Qwen2.5-3B + LoRA Adapter]
    ↓
[Empathetic Response → Gradio UI]
```

	Sarcasm Classifier	Therapist Model
Model	DistilBERT	Qwen2.5-3B + LoRA
Dataset	daniel2588/sarcasm	mental-health-chat
Data size	50k balanced samples	2,350 train / 262 test
Epochs	3	3
Training time	1.5 hours	2 hours
Hardware	Colab T4 GPU	Colab T4 GPU
Evaluation metric	Accuracy + F1	Training + validation loss

Demonstration of the AI system

DistilBERT Sarcasm Classifier

- Best Accuracy: 73.08% (Epoch 1)
- Best F1 Score: 0.7301 (Epoch 1)
- Key Finding: Mild overfitting after Epoch 1; validation loss increased while training loss decreased. Best checkpoint automatically saved at Epoch 1.
- Limitation: 73% accuracy is a realistic ceiling due to label noise, lack of contextual cues (tone, facial expressions), and inherent ambiguity of sarcastic statements.

Qwen2.5-3B LoRA Fine-Tune (Therapeutic Response)

- Final Validation Loss: 2.08 (Step 200, Epoch 3)
- Trainable Parameters: 3,686,400 / 3,089,625,088 (0.12% via LoRA)
- Key Finding: Stable generalization with consistent loss decrease (2.17 → 2.08); small gap between training and validation loss indicates no overfitting. Model learned therapist-style tone and response structure without memorizing exact outputs.
- Limitation: Dataset of 2,775 examples is relatively small for a 3B-parameter model. Language model loss alone does not capture response quality.

Project Management & Timeline:

Sprint 1 (Mar 9 - Mar 16): Selected models (Llama-8b). Downloaded and tested models locally on MacBook. Switched models due to MacBook memory constraints

Sprint 2 (Mar 16 - Mar 24): Spring Break

Sprint 3 (Mar 24 - Mar 31): Original model was gated; switched to open-source alternative

Sprint 4 (Mar 31 - Apr 1): Migrated to Google Colab due to Mac processing limitations and slow load times. Set up Colab environment with GPU access.

Sprint 6 (Apr 7 - Apr 14): Trained DistilBERT sarcasm classifier on 50k balanced samples → 73.08% accuracy, 0.7301 F1.

Sprint 6 (Apr 14 - Apr 21): Fine-tuned Qwen2.5-3B with LoRA (0.12% parameters) → 2.08 validation loss; integrated sarcasm detection + adaptive prompt pipeline.

Sprint 7 (Apr 21- Apr 28): Deployed Gradio chat UI

Work Products

- DistilBERT classifier (finalClassifier/)
- Qwen LoRA adapter (LoRAfinal/)
- Evaluation reports (docs/)
- Gradio chat interface (Colab)
- Git repo with 2 remotes (Git LFS tracked)

Key Metrics

- Total training: ~2.5 hours (T4 GPU)
- 2 models deployed end-to-end
- Real-time responses in Gradio

References:
Models:
- Qwen2.5-3B-Instruct. Alibaba Cloud. HuggingFace Hub. https://huggingface.co/Qwen/Qwen2.5-3B-Instruct
- Sanh et al. (2019). DistilBERT, a distilled version of BERT. arXiv:1910.01108

Fine-tuning & LoRA:
- Hu et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685

Libraries:
- Wolf et al. (2020). HuggingFace Transformers. arXiv:1910.03771
- HuggingFace PEFT. https://github.com/huggingface/peft
- HuggingFace TRL. https://github.com/huggingface/trl
- Dettmers et al. (2023). bitsandbytes. arXiv:2208.07339

Datasets:
- mpingale. mental-health-chat-dataset. HuggingFace Hub. https://huggingface.co/datasets/mpingale/mental-health-chat-dataset
- daniel2588. sarcasm. HuggingFace Hub. https://huggingface.co/datasets/daniel2588/sarcasm

Platform:
- Google Colab. Google Research. https://colab.research.google.com

Q & A Time